

VAIDAHI PATEL

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EDUCATION

Master of Science in Computer Science

Arizona State University, Courses: Software Security, Software Testing, Machine Learning

Jan 2024 – Dec 2025

AZ, USA

Bachelor of Technology in Computer Science and Engineering

Nirma University, Courses: Data Structures & Algorithms, Operating Systems, Object Oriented Programming

Jul 2019 – May 2023

Gujarat, India

EXPERIENCE

Software Engineer Intern

TORQ Sports

May 2025 – Aug 2025

- Developed a cross-platform **sports analytics SaaS** app using **Flutter (Dart)** for **Android/iOS**, delivering 13+ features enabling athletes and coaches to manage performance insights, **video analysis**, and social engagement for 3K+ users.
- Built a modular backend with **NodeJS**, **Express**, and **PostgreSQL**, integrating **AWS S3** for encrypted media storage, **FFmpeg** for video preprocessing, and **Firebase (Google OAuth)** for authentication.
- Implemented interactive **React** and Tailwind CSS modules for video upload/playback flows, client-side caching, pagination, and real-time feed updates, improving load times from 4.1s to 1.5s.

Full Stack Developer and Research Assistant

Indian Institute of Technology (IIT Gandhinagar)

Aug 2023 – Dec 2023

- Built a **real-time data ingestion pipeline** handling 10,000+ tweets/hour via a custom API, using **Apache Kafka** for stream processing to achieve scalable ingestion with 90% lower data loss.
- Designed a multilingual text annotation platform with **Python (FastAPI)**, **React (TypeScript and Redux)**, and **MongoDB**, optimizing database indexes and aggregations to reduce query latency by 35% on datasets with 1M+ samples.
- Containerized and deployed the application on **AWS EC2** using Docker, implementing automated **CI/CD pipelines** and test suites achieving 95% test coverage and zero-downtime deployments.

Software Developer Intern

Maxgen Technology Pvt Ltd

Jan 2023 – Jul 2023

- Created **Python**-based ETL scripts to aggregate, clean, and analyze performance data from multiple systems, generating reports for 1,200+ employees.
- Integrated **Zoho REST API** for automated data ingestion and dashboards, reducing manual entry by 40% and saving 15 hours/week.
- Deployed scripts on **AWS EC2** with **CI/CD** pipelines, logging, and secure access, improving report generation efficiency by 35%.

Software Engineer Co-op

Fortune Panel Systems

Jan 2022 – Dec 2022

- Created an **IoT-based data processing system** using **TLS-secured WebSockets** to collect telemetry (TPH, bitumen %, filler %, GPS) from 1,000+ road construction plants into **PostgreSQL**.
- Built dynamic **ASP.NET Core Razor Pages** dashboards for real-time visualization of machine performance, including search, export, and live updates via **AJAX/SignalR**.
- Optimized dashboards with server-side rendering and on-demand data fetching, enabling seamless real-time telemetry for operators and clients.

PROJECTS

Period Tracker Web App | Python, FastAPI, React, JavaScript, Supabase

- Architected a scalable full-stack AI-driven menstrual cycle tracking platform with **Supabase** and **PostgreSQL** with real-time data synchronization delivering personalized insights across 5+ user preference categories.
- Devised low-latency, 15 asynchronous REST APIs with **FastAPI** across 4 modules (authentication, preferences, AI insights, notifications), implementing JWT authentication, and integrated a **Gemini AI agent** for adaptive, context-aware recommendations.
- Designed 8+ dynamic pages (**AI chatbot**, calendar, dashboards) with reusable UI modules, global state management for 5+ user preferences, and optimized rendering using lazy loading and memoization, reducing bundle size by 28%.

Real-Time Communication System | Spring Boot, Docker, Kubernetes, Apache Kafka, JWT, WebSocket, CI/CD

- Engineered a distributed chat platform with **Spring Boot microservices**, containerized via **Docker** and orchestrated with **Kubernetes**, handling 10,000+ concurrent users.
- Developed Authentication, Messaging, Upload, and Notification services with JWT, **WebSocket**, and **Kafka**, reducing latency by 20%.
- Automated deployment with **CI/CD pipelines**, increasing deployment speed by 30% and reducing infrastructure costs by 15%.

TECHNICAL SKILLS

Languages	Python, Dart, JavaScript (ES6+), TypeScript, Java, C++, C#, SQL, HTML5/CSS3
Frameworks/Libraries	React, Angular, Node.js, FastAPI, ASP.NET Core, Flutter, Redux, Tailwind CSS
Databases	PostgreSQL, MySQL, MongoDB, Firestore
Cloud & DevOps	AWS (S3, Lambda), Azure, GCP, Docker, Kubernetes, Jenkins
Tools & Platforms	Git/GitHub, Unix/Linux, Command Line, Postman, Android Studio, Xcode
Other Skills	ETL Development, Data Visualization (React Charts, D3.js, Chart.js), Sports Analytics (Statcast Concepts)
AI Tools	Cursor, Gemini CLI, Claude Code, Github Copilot